

Pins A4 and A5 of the Arduino board are connected to DATA and CLOCK pins of the FM module (FM1), respectively. Pins 10 through 12 of the Arduino board are connected to EN, R/W and RS pins of LCD1 while pins 2 through 5 are connected to data pins of the LCD. Pins 7 and 8 of the Arduino board are connected to tactile switches S1 and S2 to increase and decrease FM frequency, respectively.

TEA5767 receiver module. The TEA5767 is a single-chip electronically-tuned FM stereo radio module for low-voltage applications with fully-integrated intermediate frequency (IF) selectivity and demodulation. The TEA5767 module is shown in Fig. 3. Its operating power supply can range from 2.5V to 5V (maximum).

The radio module is completely adjustment-free and requires a minimum of small and low-cost external components. The oscillator in TEA5767 operates with a 32.768kHz or 13MHz external crystal. The module can be configured to operate either in PC

mode or 3-wire bus mode, selectable via BUSMODE pin. In this project, the module is operated in PC mode by grounding the BUSMODE pin.

The module also provides 4-bit level information via the bus. There is autonomous search tuning function inbuilt in the module along with many more features.

Pins 3 and 6 of TEA5767 module (FM1) are connected to ground. Pins 7 and 8 of FM1 provide outputs for right and left channel, respectively.

LM386 amplifier. The output signals from the module are very low and inaudible, so we need to use an audio-amplifier circuit. Here, for testing purpose, we use low-voltage power amplifier LM386 at the left channel output. Use VR2 to adjust the volume in the speaker. If you want stereo output, you should use another LM386 with the same configuration at the right channel output of the module.

Software program

The software of this project is written in Arduino programming language. The Arduino UNO is programmed using Arduino IDE software. ATmega328 on Arduino UNO comes with a boot loader that allows you to upload new code to it without the use of external

hardware programmer. It communicates using the STK500 protocol. You can also bypass the boot loader and program the microcontroller through ICSP (in circuit serial programming) header, but using boot loader programming is quick and easy. Select the correct board from 'Tools → Board' menu in Arduino IDE and burn the program (sketch) through standard USB port in the computer.

The source code for the same is provided with auto-scan function as well as without auto-scan function.

Programming steps. The program for this circuit performs the following functions:

1. Initialising the TEA5767 module
2. Writing the data to TEA5767 module
3. Reading the TEA5767 module values
4. Displaying information on the LCD screen
5. Scanning available FM channels
6. Increasing or decreasing the frequency by a unit defined value of particular MHz.

The source program is well commented, which can be seen by opening the file in Arduino software.

Writing data to TEA5767. The useful details about writing the byte and reading the byte can be obtained from page numbers 13 through 16 of the datasheet (Rev.5).

Wire.BeginTransmission(0x60);

This is used to start PC communication with the TEA5767 module

The address of TEA5767 is 0x60 during writing

The address of TEA5767 is 0x61 during reading

Wire.write(0x00);

This is used to send data to the I²C device byte by byte

The significance of various bytes in TEA5767 is also mentioned on page numbers 13 through 16 of its datasheet. The data byte sequence for the write mode is shown in Fig. 4; the remaining detail can be found in the datasheet.

Bitwise operators used for programming the first two bytes of TEA5767 are explained below:

Test Points

Test point	Details
TP0	0V
TP1	5V
TP2	9V
TP3	Low when S1 is pressed
TP4	Low when S2 is pressed
TP5	Output of FM1

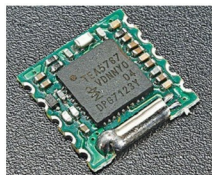


Fig. 3: TEA5767 digital radio receiver module

PARTS LIST

Semiconductors:

- FM1 - TEA5767 digital radio-receiver module
- LCD1 - 16x2 LCD
- BOARD1 - Arduino UNO board
- IC1 - LM386 low-power amplifier

Resistors (all 1/4-watt, ±5% carbon):

- R1, R2 - 10-kilo-ohm
- R3 - 330-ohm
- R4 - 10-ohm
- VR1, VR2 - 10-kilo-ohm preset

Capacitors:

- C1 - 22µF, 25V electrolytic
- C2 - 10µF, 25V electrolytic
- C3 - 100µF, 25V electrolytic

Miscellaneous:

- S1, S2 - Tactile switch
- BATT.1 - 9V battery
- LS1 - 8-ohm, 0.5W speaker
- ANT. - 75cm hookup wire antenna



Fig. 4: Write mode data sequence